ZAFEER AHMED QADEER

Mechanical Engineer | M.S. NYU Tandon '26 | Product Design · Manufacturing & Process · HVAC/MEP | SolidWorks CSWA · GD&T · Revit MEP
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PROFESSIONAL SUMMARY

Mechanical engineer with an M.S. from NYU Tandon (GPA 3.73, May 2026), working across mechanical product design, manufacturing and process engineering, and building HVAC/MEP systems. Co-founded Ingentas Robotics and delivered 8 client products to paying customers, $7,000+ in engineering revenue, every one on time. At a tooling manufacturer I traced recurring mold defects back to gaps in dimensional inspection, revised those checkpoints, and cut rework 10-15%. Comfortable from concept and CAD through tolerance analysis, simulation, process selection, fabrication and acceptance testing. First-author publication on LiDAR and AI in drones, IOSR-JMCE, April 2024. Authorized to work in the U.S.

TECHNICAL SKILLS

CAD & Mechanical Design: SolidWorks (CSWA), AutoCAD, Fusion 360, CATIA, Solid & Surface Modeling, Large Assembly Modeling, GD&T (ASME Y14.5), Tolerance Stack-up Analysis, DFM, DFA, DFMEA, Sheet Metal Design, Engineering Drawings, BOM Management

Manufacturing & Fabrication: Injection Molding, Blow Molding, Die Casting, Rotational Molding, CNC Machining, Additive Manufacturing, FDM, SLA, Rapid Prototyping, Hand Prototyping, Tooling Design, Fixture Design, Material Selection, Vendor Sourcing, Low-Volume Manufacturing

Process & Quality: Process Optimization, Root Cause Analysis, Continuous Improvement, Dimensional Inspection, Standard Work Instructions, Statistical Process Control, FMEA, Lean Manufacturing, 5S, Poka-Yoke, ISO 9001, Design Validation, Takt Time, QA Documentation

Simulation & Analysis: ANSYS (Structural and Thermal FEA), SolidWorks Flow Simulation (certified), Computational Fluid Dynamics, MATLAB, Thermal Analysis, Heat Transfer, Fluid Mechanics, Kinematic Analysis, Statics

Electromechanical & Embedded: Sensor Integration (LiDAR, ultrasonic, camera, FPV), Motor, ESC and Controller Packaging, Wiring and Cable Routing, Arduino (Uno, MEGA), Firmware Configuration, C, C++

HVAC & MEP Systems: HVAC System Design, Cooling & Heating Load Calculations, Duct & Pipe Sizing, Psychrometrics, VRF/VRV Systems, MEP Coordination, Revit MEP, AutoCAD MEP, Construction Documents, Commissioning, As-Built Verification

Codes & Standards: ASHRAE 62.1, ASHRAE 90.1, IMC, NFPA 90A/90B, NYCECC, Local Law 97

EDUCATION

New York University - Tandon School of Engineering May 2026
M.S. Mechanical Engineering | GPA 3.73/4.0 | New York, NY
Coursework: New Product Development, Additive Manufacturing of Metallic Materials, Security in Additive Manufacturing, CFD & Heat Transfer, Thermal Engineering Fundamentals, Energy Project Financing, Linear Control Theory, Solid Mechanics

Muffakham Jah College of Engineering & Technology (Osmania University) Jun 2024
B.E. Mechanical Engineering | Hyderabad, India
Coursework: Design for Manufacturing, Mechanics of Materials, Thermal Engineering, Fluid Mechanics

PROFESSIONAL EXPERIENCE

NYU Hyperloop Team, NYU Tandon Jan 2026 – May 2026
Mechanical Design Engineer | New York, NY
- Held mechanical routing and installation clearances against the electrical and structural subsystems on a 20+ member team, resolving the interferences in design reviews before anything reached fabrication.
- Applied GD&T (ASME Y14.5) and DFM across 30+ custom pod components, releasing drawings cleared for fabrication.
- Maintained the engineering BOM, assembly drawings and design rationale the team worked from across sprint reviews.

Sun Max Solar Systems Oct 2023 – Jan 2024
Solar Energy Engineering Intern | Hyderabad, India
- Ran site surveys and commissioning walkthroughs across residential and commercial PV installations, checking as-built conditions against design drawings and writing the handover documentation clients signed off.
- Performed QA on lithium-ion and lead-acid battery systems and coordinated installations against local regulatory requirements.

Advance Tooling Systems Jun 2023 – Jul 2023
Manufacturing Engineering Intern, Tooling & Manufacturing | Hyderabad, India
- Traced recurring mold fitment defects back to gaps in dimensional inspection and revised those checkpoints, cutting rework 10-15%, saving 1.5-2 hours per production cycle, and improving quality on customer consignments by 40%.
- Selected polymers and alloys for mold fabrication across 50+ tooling units spanning injection molding, blow molding, die casting and rotational molding, and evaluated draft, wall thickness and tooling feasibility on each.
- Kept the QA records for every unit inspected against supplied blueprints.

Mahindra & Mahindra Limited (Global Automotive OEM) May 2023 – Jun 2023
Manufacturing Engineering Intern, Automotive & Farm Equipment | Zaheerabad, India
- Mapped a multi-stage tractor assembly line, documenting station-level quality gates, takt time and material flow for the plant engineering team.

Ingentas Robotics (Engineering Startup) Jan 2022 – Sep 2022
Mechanical Design & Manufacturing Engineer | Co-Founder | Hyderabad, India

- Co-founded the firm and took 8 products to paying customers: a UAV quadcopter, a Bluetooth-controlled RC vehicle, an ultrasonic-sensing vehicle and a line-following autonomous vehicle. $7,000+ in engineering revenue over 9 months, every one delivered on time.
- Packaged motors, ESCs, flight controllers, sensor modules and wiring into compact 3D-printed and CNC-machined chassis, then ran fit, clearance and acceptance testing across successive prototype builds.
- Sourced from outside vendors on small volumes and short lead times, trained the build team, and explained design decisions to customers who were not engineers.

ENGINEERING PROJECTS

Motorcycle CAD Study - SolidWorks Jul 2026 – Aug 2026
- Modeled a 30-part motorcycle assembly in SolidWorks from open-source reference drawings: weldment frame, surfaced 2 mm thin-wall fuel tank, cast crankcase cover with draft, and 3 subassemblies mated into a top-level assembly.

SolidWorks Flow Simulation Portfolio - Five Applied CFD Studies Aug 2026
- Venturi discharge coefficient validated against ISO 5167-4 within 1.16%. Counter-current heat exchanger LMTD hand-verified to 0.007%. Ventilation CO2 decay time constant within 0.83% of single-zone theory. Seven independent validations across five problems.

Design and Fabrication of Stair Climbing Robot - B.E. Capstone, Team of 3 2024
- Designed the belt drive, pulleys and 6061-T6 aluminum frame in SolidWorks, sourced the bill of materials, and built the machine. Tested across four stair geometries measuring speed, power draw and stability, and traced the failure at a 45-degree rise to traction rather than motor torque, since the statics showed 2.4x margin at that point.

TerraExplorer - Six-Wheel Remote Sensing Rover | MJCET R&D Cell Oct 2023 – May 2024
- Only mechanical engineer on a four-person electronics team. Designed the chassis and rocker-bogie suspension in SolidWorks around 20x20mm aluminum T-slot profile and 3D-printed joints, then printed and assembled the running vehicle carrying an FPV camera and RF link. Received the R&D Cell's Letter of Appreciation.

Kinematic Gear Drive Vehicle (KGDV) | NYU Tandon May 2026
- Designed a fully 3D-printed multi-stage gear drivetrain at a 3:1 overall ratio, ran worst-case tolerance stack-up across the 3-stage train, specified +/-0.15mm running clearances to absorb FDM dimensional variation, and iterated 3 print cycles to a working 20-part assembly.

Solar Desalination Plant - CSP + Multi-Effect Distillation | NYU Graduate Project Mar 2025 – May 2025
- Designed a 500,000 L/day plant around a CSP power tower with 8-hour molten salt storage, ran the thermal calculations across the solar field, storage and MED train, and costed it at $27M CAPEX against a 10-12 year break-even.

Commercial Retrofit HVAC Energy & Cost Analysis | NYU Coursework May 2026
- Applied lifecycle cost modeling to HVAC upgrade scenarios for a commercial building, comparing energy use, installation cost and Local Law 97 penalty exposure across a 15-year horizon to find the lower-NPV compliance path.

SAE Team Asphalt (Captain) | Club Optimus (Founder & Chairman), MJCET May 2021 – Jul 2024
- Finished 2nd nationally at SAE IPEC 2K23 leading a 15+ member team through design, fabrication and testing of an electric go-kart. Ran Ackermann steering geometry, braking load and structural thermal analysis in ANSYS and SolidWorks for SAE BAJA 2022. Grew Club Optimus from nothing to 100+ members across 25+ technical events.

CERTIFICATIONS & PUBLICATION

SOLIDWORKS CAD Design Associate (CSWA) - Dassault Systemes, Sep 2024
SOLIDWORKS Additive Manufacturing Associate (CSWA-AM) - Dassault Systemes, Sep 2024
SOLIDWORKS Fluid Dynamics Analysis CFD - Udemy, Omar Koryakin, Aug 2026
Mastering SOLIDWORKS (2024-25): A Complete Course - Udemy, CADCIM Technologies, Aug 2026
HVAC Troubleshooting PRO: HVAC Maintenance & Diagnosis - Udemy, ExpertEase Global, Jul 2026
Publication (first author): "Advancement in Drone Technology Integrated With Artificial Intelligence and LiDAR Sensor", IOSR Journal of Mechanical and Civil Engineering, Vol. 21(2), pp. 35-41, April 2024